

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development and Jira

Practical – I

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AIM : Introduction to AI and Intelligent Agents

Design a Simple Rational Agent by defining PEAS for Vacuum Cleaner Environment and Autonomous Taxi. Classify environments (fully/partially observable, deterministic/stochastic, episodic/sequential). Implement a simple table-driven agent in Python.

PEAS: P:Performance , E: Environment , A:Actuators , S:Sensor

CODE: Vacuum Cleaner Environment

```
table = {
    ('A' , 'Dirty'): 'Suck',
    ('A' , 'Clean'): ' Move Right',

    ('B' , 'Dirty'): 'Suck',
    ('B' , 'Clean'): 'Move Forward',

    ('C' , 'Dirty'): 'Suck',
    ('C' , 'Clean'): 'Move Backward',

    ('D' , 'Dirty'): 'Suck',
    ('D' , 'Clean'): 'Move Left'
}

sequence_table = {
    (('A' , 'Dirty') , ('A' , 'Clean')): 'Move Right',
    (('A' , 'Clean') , ('A' , 'Dirty')): 'Suck',
    (('B' , 'Dirty') , ('B' , 'Clean')): 'Move Forward',
    (('B' , 'Clean') , ('B' , 'Dirty')): 'Suck',
    (('C' , 'Dirty') , ('C' , 'Clean')): 'Move Backward',
    (('C' , 'Clean') , ('C' , 'Dirty')): 'Suck',
    (('D' , 'Dirty') , ('D' , 'Clean')): 'Move Left',
```

```
(('D' , 'Clean') , ('D' , 'Dirty')): 'Suck'
}

percepts = []

def table_driven_agent(percept):
    percepts.append(percept)

    if tuple(percepts) in sequence_table:
        return sequence_table[tuple(percepts)]

    return table.get(percept , "No Operation")

print("Vaccum Cleaner Table-Driven Agent")

n = int(input("Enter no.of Percepts: "))

for i in range(n):
    print(f"\nPercept {i + 1}")

    location = input("Enter Location (A/B/C/D): ").upper()
    status = input("Enter Status (Clean/Dirty): ").capitalize()

    percept = (location , status)
    action = table_driven_agent(percept)

    print("Percept History: ",percepts)
    print("Agent Action: ",action)
```

OUTPUT:

Vaccum Cleaner Table-Driven Agent

Percept 1

Percept History: [('A', 'Dirty')]

Agent Action: Suck

Percept 2

Percept History: [('A', 'Dirty'), ('B', 'Clean')]

Agent Action: Move Forward

Percept 3

Percept History: [('A', 'Dirty'), ('B', 'Clean'), ('C', 'Dirty')]

Agent Action: Suck

Percept 4

Percept History: [('A', 'Dirty'), ('B', 'Clean'), ('C', 'Dirty'), ('C', 'Clean')]

Agent Action: Move Backward

Percept 5

Percept History: [('A', 'Dirty'), ('B', 'Clean'), ('C', 'Dirty'), ('C', 'Clean'), ('D', 'Dirty')]

Agent Action: Suck

CODE: Autonomous Taxi

```
# (Location, Traffic Signal, Passenger in Car, Ride Request)
```

```
table = {  
    ('A', 'Green', 'No', 'Yes'): 'Pick Up Passenger',  
    ('A', 'Green', 'No', 'No'): 'Move Forward',  
  
    ('B', 'Red', 'Yes', 'No'): 'Stop',  
    ('B', 'Green', 'Yes', 'No'): 'Drive to Destination',  
  
    ('C', 'Yellow', 'No', 'No'): 'Slow Down',  
    ('C', 'Green', 'No', 'No'): 'Pick Up Passenger',  
  
    ('D', 'Red', 'Yes', 'No'): 'Stop',  
    ('D', 'Green', 'Yes', 'Yes'): 'Drop Passenger'  
}
```

```
percepts = []
```

```
def taxi_agent(percept):  
    percepts.append(percept)  
    action = table.get(percept, "Wait")  
    return action
```

```
print("Autonomous Taxi Table-Driven Agent")

n = int(input("Enter the no.of percepts: "))

for i in range(n):
    print(f"\nPercept {i + 1}")

    location = input("Location (A/B/C/D): ").upper()
    signal = input("Traffic Signal (Green/Yellow/Red): ").capitalize()
    passenger = input("Passenger in Car? (Yes/No): ").capitalize()
    request = input("Ride Request? (Yes/No): ").capitalize()

    percept = (location, signal, passenger, request)

    action = taxi_agent(percept)

    print("Current Percept:", percept)
    print("Agent Action:", action)
```

OUTPUT:

Autonomous Taxi Table-Driven Agent

Percept 1

Current Percept: ('A', 'Green', 'No', 'Yes')

Agent Action: Pick Up Passenger

Percept 2

Current Percept: ('A', 'Green', 'No', 'No')

Agent Action: Move Forward

Percept 3

Current Percept: ('B', 'Red', 'Yes', 'No')

Agent Action: Stop

Percept 4

Current Percept: ('B', 'Green', 'Yes', 'No')

Agent Action: Drive to Destination

Percept 5

Current Percept: ('C', 'Yellow', 'No', 'No')

Agent Action: Slow Down

Percept 6

Current Percept: ('C', 'Green', 'No', 'No')

Agent Action: Pick Up Passenger

Percept 7

Current Percept: ('D', 'Red', 'Yes', 'No')

Agent Action: Stop

Percept 8

Current Percept: ('D', 'Green', 'Yes', 'Yes')

Agent Action: Drop Passenger

CODE: Own

```
# (Location, Temperature, Motion Detected)

table = {
    ('Living Room', 'Hot', 'Yes'): 'Turn On AC',
    ('Living Room', 'Cold', 'No'): 'Turn Off AC',

    ('Bedroom', 'Hot', 'Yes'): 'Turn On Fan',
    ('Bedroom', 'Cold', 'No'): 'Turn Off Fan',

    ('Kitchen', 'Hot', 'Yes'): 'Start Exhaust Fan',
    ('Kitchen', 'Cold', 'No'): 'Turn Off Exhaust Fan',

    ('Garage', 'Cold', 'Yes'): 'Turn On Light',
    ('Garage', 'Hot', 'No'): 'Turn Off Light'
}

percepts = []

def smart_home_agent(percept):
    percepts.append(percept)
    return table.get(percept, "No Action")

print("Smart Home Table-Driven Agent")

n = int(input("Enter number of percepts: "))

for i in range(n):
    print(f"\nPercept {i + 1}")

    location = input("Enter Location (Living Room/Bedroom/Kitchen/Garage): ").title()
    temperature = input("Enter Temperature (Hot/Cold): ").capitalize()
    motion = input("Motion Detected? (Yes/No): ").capitalize()

    percept = (location, temperature, motion)

    action = smart_home_agent(percept)

    print("Current Percept :", percept)
    print("Percept History :", percepts)
    print("Agent Action  :", action)
```

OUTPUT:

Smart Home Table-Driven Agent

Percept 1

Current Percept : ('Living Room', 'Hot', 'Yes')

Percept History : [('Living Room', 'Hot', 'Yes')]

Agent Action : Turn On AC

Percept 2

Current Percept : ('Bedroom', 'Cold', 'No')

Percept History : [('Living Room', 'Hot', 'Yes'), ('Bedroom', 'Cold', 'No')]

Agent Action : Turn Off Fan

Percept 3

Current Percept : ('Kitchen', 'Hot', 'Yes')

Percept History : [('Living Room', 'Hot', 'Yes'), ('Bedroom', 'Cold', 'No'), ('Kitchen', 'Hot', 'Yes')]

Agent Action : Start Exhaust Fan

Percept 4

Current Percept : ('Garage', 'Cold', 'No')

Percept History : [('Living Room', 'Hot', 'Yes'), ('Bedroom', 'Cold', 'No'), ('Kitchen', 'Hot', 'Yes'), ('Garage', 'Cold', 'No')]

Agent Action : No Action